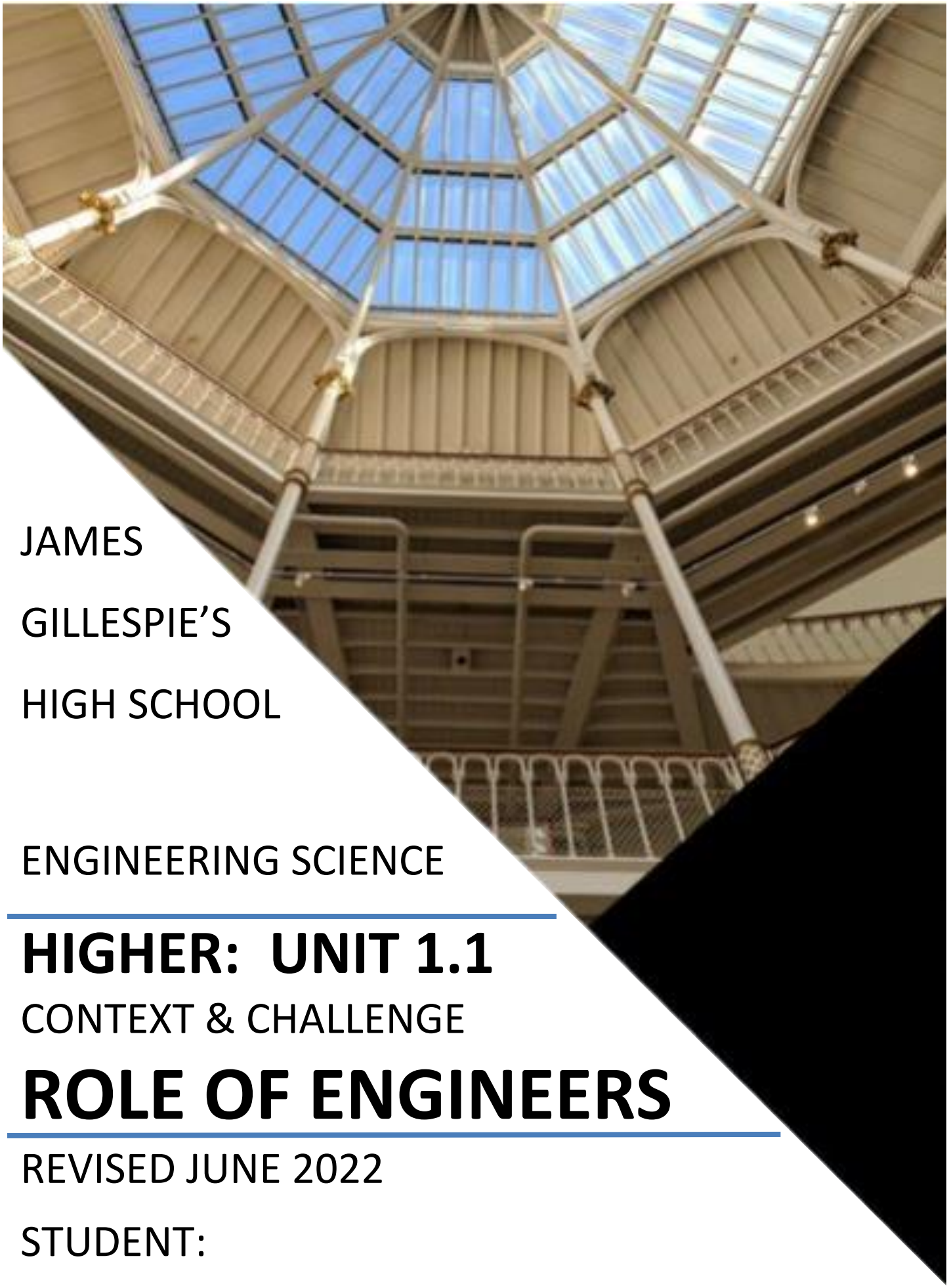


JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 1.1

CONTEXT & CHALLENGE

ROLE OF ENGINEERS

REVISED JUNE 2022

STUDENT:

YEAR: TEACHER:

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Don't forget

There is a big step from National 5 to Higher and your answers are expected to demonstrate this. Answers should never be a simple statement. A significant portion of the marks gained in the Higher question paper will be from your descriptive answers. If you are asked about engineering role in an exam, it should be of Higher English level quality and should describe. This means talking about the cause of something and the effect it then has.

Outcomes and Assessment Standards

The aim of this unit is to develop a deeper understanding of engineering, its role and impact on our society and environment. There are main 3 outcomes in this unit and it has been divided into three booklets. By the end of book 1.1, you will have completed only outcome 3. You will investigate engineering systems, problems and solutions, involving some existing and emerging technologies and consider implications relating to the environment, sustainable developments, and to economic and social issues.

Outcome 1

1 Research and describe a complex engineering system by:

1.1 Analysing the needs being met by the system.
1.2 Identifying sub-systems, and describing the function of each and how they interact.
1.3 Producing system and sub-system diagrams.
1.4 Explaining the role of feedback in the system.
1.5 Carrying out an energy audit of the system.

The system researched should include both mechanical and electronic aspects.

Outcome 2

2 Model aspects of a complex engineered solution by:

2.1 Constructing or simulating a model of its control system.
2.2 Constructing or simulating a model of a mechanical or structural aspect of the solution.

Outcome 3 (book 1.1)

3 Present a critical analysis of an engineered solution to a contemporary problem by:

3.1 Describing clearly the nature of the problem.
3.2 Describing some social and economic impacts of the solution.
3.3 Describing clearly some environmental impacts of the solution.
3.4 Identifying and describing emerging technologies which may impact future developments.

Skills, knowledge and understanding for the course assessment

The following provides details of skills, knowledge and understanding sampled in the course assessment. Place a tick beside each topic once you have been taught it to track your progress of learning. You might want to use traffic lighting to record your Knowledge and Understanding. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will regularly review this record page after speaking to you about your self-evaluation.

Red: I need to work on this topic further to understand it.

Amber: I need a bit of help to fully understand this topic.

Green: I am confident in my understanding of this topic.

- ☐ I understand the role of various engineers within design projects.
- ☐ I can identify the differences between skills and knowledge used by a variety of engineering disciplines.
- ☐ I am Identifying and describing emerging technologies which may impact future developments.
- ☐ I can describe some social and economic impacts of the solution.
- ☐ I can describe clearly some environmental impacts of the solution.
- ☐ I can identifying and describe emerging technologies which may impact future developments.
- ☐ I am able to discuss possible lifetime costs of the environment in Scotland and beyond.

Judging evidence for Engineering Contexts and Challenges (Higher)

Outcomes	Assessment Standards	Making assessment judgements	Assessment for candidates (Appendix 1): commentary on assessment judgements
1 **Research and describe a complex engineering system by:**	1.1 Analysing the needs being met by the system	The candidate identifies all inputs and outputs for the overall system.	The complex engineering system chosen must involve closed loop control, and should include both mechanical and electronic aspects. All three Outcomes may be based on the same engineering system / solution, or each Outcome may be based on a different engineering system / solution. Explanations may be written or oral. Observational evidence, supported by photograph(s) of models or hard copy from simulations, with assessor comments. Partial exemplar evidence is illustrated on pages 9-10, based on an intelligent wiper system.
	1.2 Identifying sub-systems, and describing the function of each and how they interact 1.3 Producing system and sub-system diagrams	Diagrams should show: ♦ control unit ♦ driver(s) ♦ output transducer(s) ♦ sensor(s) ♦ feedback loop ♦ system boundary	
	1.4 Explaining the role of feedback in the system	Candidate must state the type of control (eg two-state or proportional), and explain how it is implemented.	
	1.5 Carrying out an energy audit of the system	Energy audit should include: ♦ calculation of energy/power inputs and/or outputs ♦ comments on possible improvements in efficiency	
2 **Model aspects of a complex engineered solution by:**	2.1 Constructing or simulating a model of its control system	Working model or simulation.	Observational evidence, supported by photograph(s) of models or hard copy from simulations, with assessor comments. Partial exemplar evidence on pages 10-11, shows good practice in control system programming.
	2.2 Constructing or simulating a model of a mechanical or structural aspect of the solution	Working model or simulation.	

Outcomes	Assessment Standards	Making assessment judgements	Assessment for candidates (Appendix 1): commentary on assessment judgements
3 **Present a critical analysis of an engineered solution to a contemporary problem by:**	3.1 Describing clearly the nature of the problem	Description clearly outlines the problem.	The evidence may be any appropriate format, such as a written report or presentation. Example annotated extracts from a written report relating to wind energy as a possible solution to climate change. Assessors should ensure that the candidate selects a solution and problem which will give scope to achieve all Assessment Standards.
	3.2 Describing some social and economic impacts of the solution	The candidate produces a balanced description of the social **and** economic impact of the solution, referring to both benefits and disadvantages. Impacts must be specific to the problem solution.	
	3.3 Describing clearly some environmental impacts of the solution	The candidate produces a balanced description of the environmental impact of the solution, referring to both benefits and disadvantages. Impacts must be specific to the problem solution.	
	3.4 Identifying and describing emerging technologies which may impact future developments	Brief description of at least two emerging technologies which may be relevant to the specific solution.	

There are **three** tasks to be completed in this assessment, one for each Outcome.

- **Task 1** requires you to research and describe a complex engineering system. (book 1.2)
- **Task 2** involves constructing or simulating model aspects of a complex engineered solution. (book 1.2)
- **Task 3** requires you to present a critical analysis of an engineered solution to a contemporary problem.

You may base all of these tasks on the same system/solution, or you may choose a different system/solution for each task. Your assessor may guide you in your choice of system/solution.

The evidence for each task may be presented in any format. For example, you could produce items of written evidence, or you may choose to present your findings orally using presentation software, or a combination of the above.

Task 1 (Outcome 1) Research and describe a complex engineering system

Choose a complex engineering system from the list provided by your assessor, or alternatively a complex engineering system of your own choosing which must be discussed with your assessor prior to starting the task.

You are to research the system, and produce a description of it which will include the following:

- An analysis of the needs being met by the system: what does it do? How does it achieve this? What key inputs and outputs does the system have?
- Annotated system and sub-system diagrams which show key inputs, outputs and interconnections between the main sub-systems, describe the function of each and how they interact with each other.
- A detailed description of how the system works, explaining the role of feedback within the system.
- An energy audit of the system, identifying the main energy transformations and losses within the system. Your audit should include calculations of efficiency.

Task 2 (Outcome 2) Model aspects of a complex engineered solution.

2.1 Constructing or simulating a model of its control system

a) Choose a complex engineering system, either from a list provided by your assessor or a system of your own choice. If choosing the system yourself, this should be discussed with your assessor prior to starting the task.

b) Construct or simulate the control system. If you are using a simulation package, attach a printout of your solution. If constructing, obtain a photograph of your constructed solution.

2.2 Constructing or simulating a model of a mechanical or structural aspect of the solution

Construct or simulate a mechanical or structural aspect of the solution. If you are using a simulation package, attach a printout of your solution. If constructing, obtain a photograph of your constructed solution.

Task 3 (Outcome 3) Present a critical analysis of an engineered solution to a contemporary problem

3.1 Describing clearly the nature of the problem

In this section you should outline the nature of a recent or current problem and use examples to highlight the problem.

3.2 Describing some social and economic impacts of the solution

Describe at least one positive and one negative impact, and at least one social and one economic impact. These impacts must be specific to the problem solution.

3.3 Describing clearly some environmental impacts of the solution

Describe at least one positive and one negative environmental impact. These impacts must be specific to the problem solution.

3.4 Identifying and describing emerging technologies which may impact future developments

Why engineering context and challenge?

Throughout Higher Engineering Science, you will learn about many new and unfamiliar engineering concepts. However, these concepts do not mean much without a context which they can be applied to. Whether studying after school or moving on to pursue engineering as a career, all of the concepts learned in this course must be applied to an unfamiliar context.

Every single day we are surrounded by new and emerging technology which engineers have developed. These technologies have been designed as solutions to a problem. It is therefore important that you are able to use your understanding to and demonstrate it in relation to the outside world.

In March 2014, a report by the *Financial Times* found that 80% of engineering companies is experiencing recruitment problems. There are literally thousands of jobs to be filled in the UK every year.

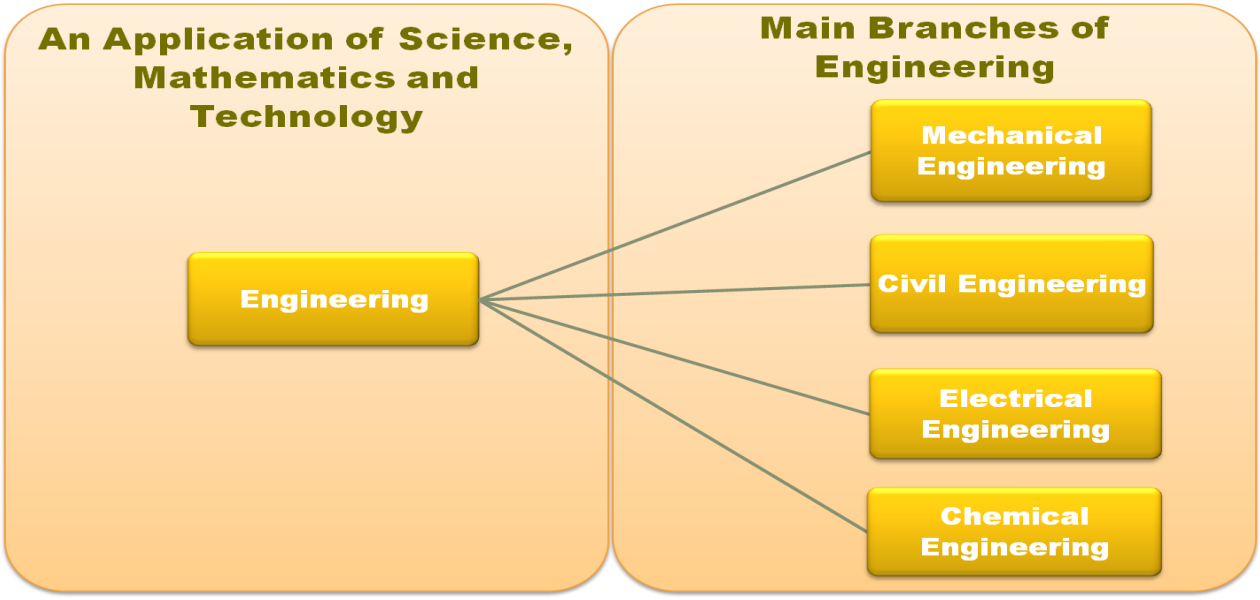
This unit will build on skills learned in National 5 Engineering Science in order to further deepen your understanding of engineering Contexts and Challenges.

What is Engineering?

We take the term 'technology' to mean the things or products made by our society and Engineers are involves in this at every stage through design, development manufacture and construction to development.

In our modern world there are so many new products being created that it is impossible for us to keep up and know everything about these developments. As Engineers we need some help to use and understand all this technology. The strategy we use is called the *top-down* approach.

At the very top of the approach we limit the study of a product or process to what we need to know. For example, to use a toaster all we need to know is that if we plug it in, set the temperature, switch it on and insert a slice of bread then, after a few minutes, we will get toast. However, engineers designing a new toaster will need to know a lot more and will have an understanding of all the parts that go to make up the toaster and how they relate to one another. In even a simple engineering project like designing a new toaster the knowledge and skills of several different engineers will be needed. The diagram below shows the main branches of Engineering.



Technology has advanced so much in recent years that Engineers must not only consider what the whole project must achieve but also consider the tiniest details. To do this requires knowledge, expertise and skills in many different areas. Consequently, Engineering is now divided into many branches. Within the Higher course, you will focus on **seven** different yet intertwined disciplines. The diagrams below show some of the Engineering types which have developed from the four main branches.

Mechanical, Civil, Electrical, Chemical, Structural, Electronic, Environmental

Remember there is a difference between knowledge and Skills *and you may be asked about these in the exam. A* ***skill*** *that an engineer has could be the ability to* ***design, calculate, simulate, analyse, justify*** *or* ***evaluate*** *part of an engineering project.* ***Knowledge*** *is what they need to know to allow this skill to be used.*

SQA past paper 2018 Q3

The engineer responsible for the structural design of a large metal sculpture chose to use alloy steel for the inner support structure, and stainless steel for the cladding. Describe two examples of specialist skills and two examples of specialist knowledge that a structural engineer would use. (Marks: 4)

Answers: A suitable answer must be able to explain and demonstrate to the SQA that you know exactly the difference between a Knowledge and a Skill for each types of engineer. An answer may contain a response which is based upon the following statements.

Knowledge

(two responses from):

- Material properties.
- Material costs.
- Available material sections.
- Using values to influence design decisions.

Skills

(two responses from): (2 marks for skills. Only 1 mark available for each skill ie. Calculating, analysing, simulating, justifying and evaluating).

- Use material values (eg Young's Modulus, UTS) in structural calculations.
- Analyse results from materials test data.
- Simulate material/structural performance using computer software.
- Justify material choices as appropriate for given situation.
- Use terminology such as 'Design, organise, implement, control, supervise, quality assurance'.
- Evaluate material performance.

Mechanical engineer role overview

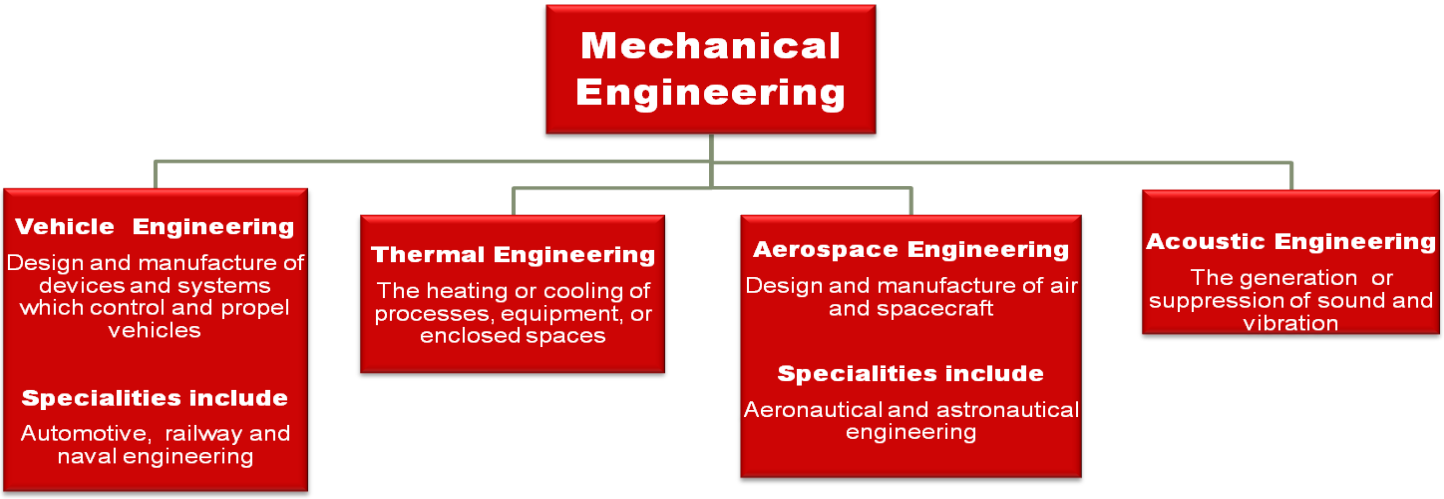
Mechanical engineers provide efficient solutions to the development of processes and products, ranging from small component designs to extremely large plant, machinery or vehicles. They can work on all stages of a product, from research and development to design and manufacture, through to installation and final commissioning. Most industries rely on a form of mechanical systems and mechanical engineering is thought to be one of the most diverse of all engineering disciplines. Due to this, there are employment opportunities in a wide range of sectors, including: Manufacturing; Power; Construction; Medical.

Knowledge and responsibilities

- Designing and implementing cost-effective equipment modifications to help improve safety and reliability.
- Developing a project specification with colleagues, often including those from other engineering disciplines.
- Developing, testing and evaluating theoretical designs.
- Discussing and solving complex problems with manufacturing departments, subcontractors, suppliers and customers.
- Making sure a product can be made reliably and will perform consistently in specified operating environments.
- Managing projects using engineering principles and techniques.
- Planning and designing new production processes.
- Producing details of specifications and outline designs.
- Recommending modifications following prototype test results; using research, analytical, conceptual and planning skills, particularly mathematical modelling and computer-aided design.
- Considering the implications of issues such as cost, safety and time constraints.
- Working with other professionals, within and outside the engineering sector.
- Monitoring and commissioning plant and systems.

Skills

- A high level of technical and scientific knowledge and the ability to apply this knowledge to practical problems.
- Good oral communication skills and confidence in dealing with a wide range of people, including clients, contractors, designers, directors and plant operators.
- Precise and concise written communication skills; the ability to work well within a team.
- IT skills, particularly computer-aided design.
- Creative ability.
- The ability to work under pressure.
- Organisational skills, such as time and resource planning.



1a In your notebook, take the heading "Mechanical Engineers - Knowledge, skills and roles" on a new page. Neatly copy the diagram above. Within each main branch of engineering, there are many specialisms. Investigate if you can find more than the ones above and record what its specialities are.

1b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Mechanical Engineers' Knowledge - What they know" and a list of "Mechanical Engineers Skills - What they can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

Civil engineer role overview

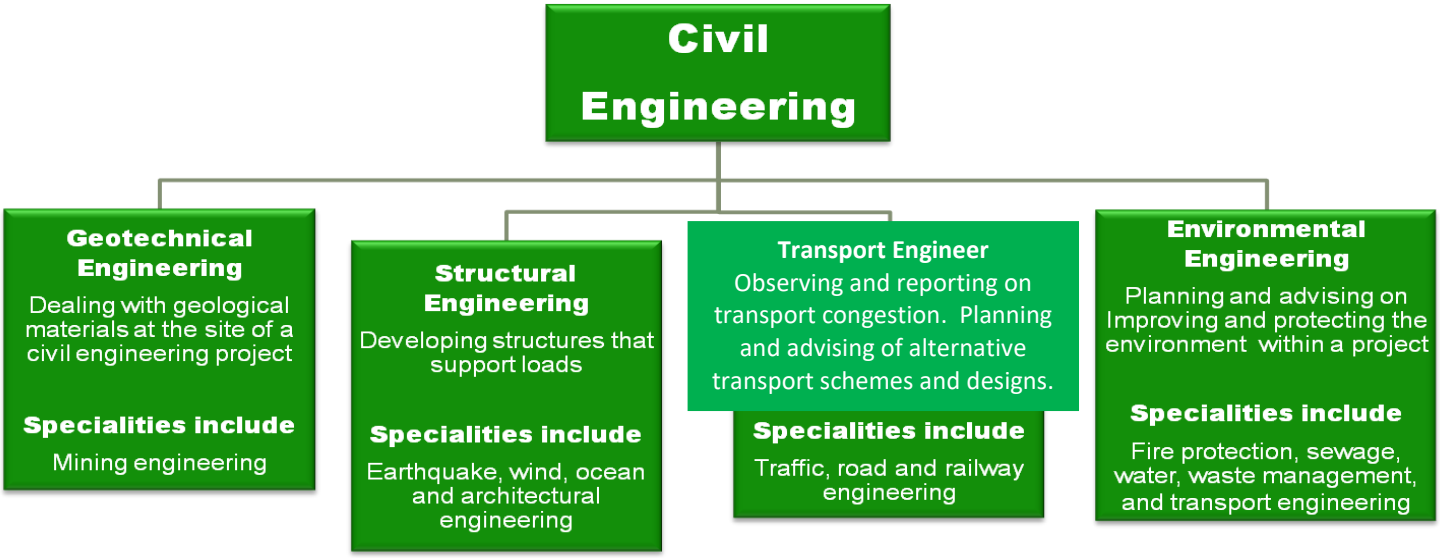
Civil engineers specialize in road, bridge, buildings and water supply system design and construction. They supervise and direct construction teams and work with other engineers. These professionals ensure that every structure built is environmentally compliant and can withstand earthquakes and hurricanes. This is especially true in places where these natural calamities often strike. Civil engineers work wherever there is a need for expanding new structures or transportation systems and geotechnical engineering. They also hold administrative and supervisory positions.

Knowledge and responsibilities

- Different forces and their behaviour and characteristics.
- How construction materials react to forces and support loads.
- What work has to be done on the ground to enable construction projects to be started.

Skills

- Design foundations for buildings and structures to be built upon.
- Design systems to prevent the natural environment affecting construction projects (e.g. flood risk reduction).
- Calculate forces in structures to ensure they are strong and don't collapse.



2a In your notebook, take the heading "Civil Engineers - Knowledge, skills and roles" on a new page. Neatly copy the diagram above. Within each main branch of engineering, there are many specialisms. Investigate if you can find more than the ones above and record what its specialities are.

2b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Civil Engineers' Knowledge - What they know" and a list of "Civil Engineers Skills - What they can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

Electrical engineer role overview

Electrical engineers design, develop and maintain electrical control systems and components to required specifications. Their work focuses on: economy; quality; reliability; safety; sustainability. The electrical equipment that they design and manufacture is used across many sectors, including: the building industry and services, including lighting, heating and ventilation; transportation and transport networks; manufacturing and construction; production and distribution of power.

Electrical engineers are involved in projects from the concept and detail of the design, through to implementation, testing and handover. They may be involved in maintenance programmes. As well as having technical knowledge, electrical engineers need to be able to project manage and multitask. They also need to have commercial awareness. Additional attributes, such as team leadership or management skills, are required as careers progress.

Knowledge and responsibilities

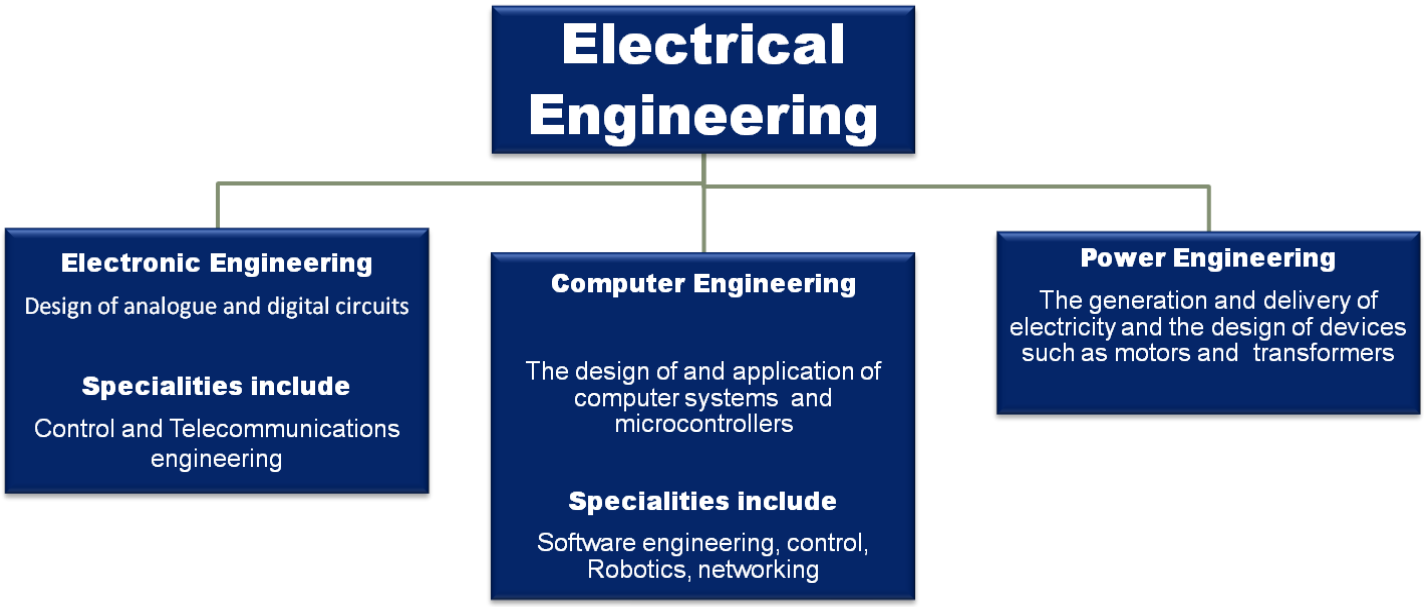
- Identifying customer requirements by conducting an audit to calculate electrical values and select correct components.
- Designing/write computer code systems and products to develop automated electronic systems.
- Reading design specifications and technical drawings.
- Researching suitable solutions and estimating costs and timescales.
- Making models and prototypes of products using three-dimensional design software; working to British

(BS), European (EN) and other standards.

- Liaising with others in the design team.
- Communicating with clients and contractors.
- Attending meetings on site.
- Designing and conducting tests of circuits to control systems and deliver energy to drive outputs.
- Recording, analysing and interpreting test data.
- Proposing modifications and retesting products.
- Qualifying the final product or system.
- Servicing and maintaining equipment.
- Preparing product documentation, writing reports and giving presentations.
-
- Monitoring a product in use to improve on future design.

Skills

- Commercial awareness.
- Oral and written communication skills.
- An enthusiasm for your subject and up-to-date sector knowledge.
- Planning and organisational skills, such as time and resource allocation.
- The ability to work in a multidisciplinary team.
- A second language could be useful when applying to international companies.



3a In your notebook, take the heading "Electrical Engineers - Knowledge, skills and roles" on a new page. Neatly copy the diagram above. Within each main branch of engineering, there are many specialists. Investigate if you can find more than the ones above and record what its specialities are.

3b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Electrical Engineers' Knowledge - What they know" and a list of "Electrical Engineers Skills - What the can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

Chemical engineer role overview

A chemical engineer designs and develops the processes that make a diverse range of products. Their work focuses on changing the chemical, biochemical and physical state of a substance to turn it into something else, for example making plastic from oil. Chemical engineers develop a knowledge and understanding of different chemical and their properties and characteristics when mixed and the reactions and processes. They understand how to alter raw materials into required products while taking into consideration health and safety and cost issues. They work in a variety of industries including oil and gas; pharmaceuticals; energy; water treatment; food and drink; plastics; toiletries.

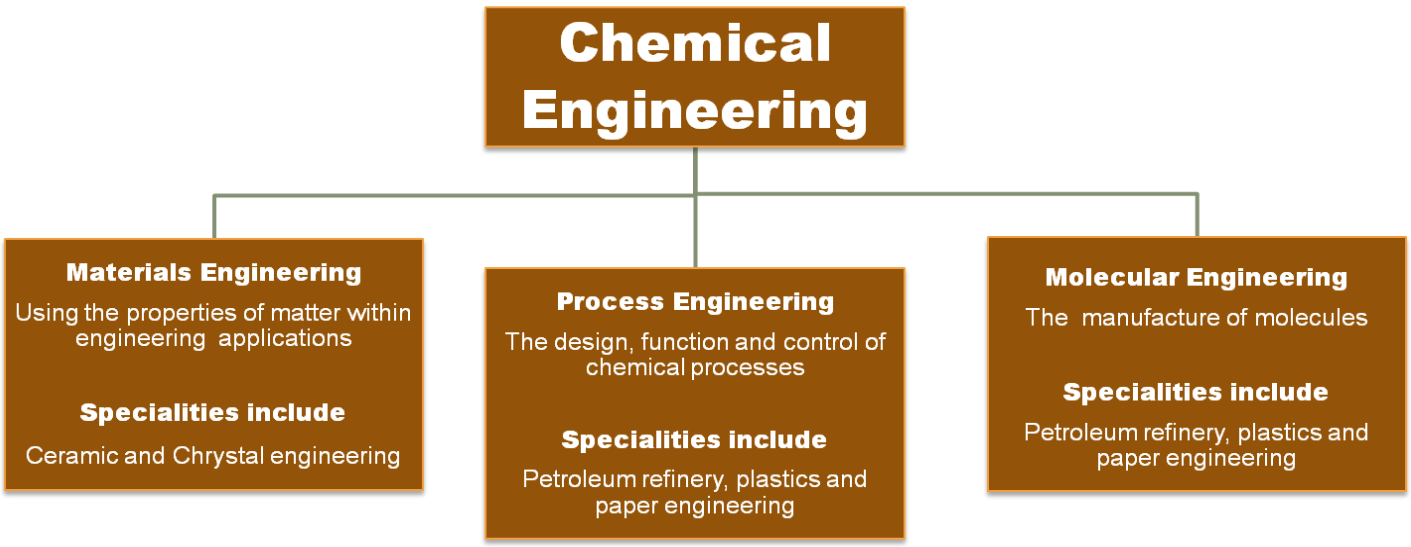
Modern chemical engineering is also concerned with pioneering valuable new materials and techniques, such as nanotechnology, fuel cells and biomedical engineering.

Knowledge and responsibilities

- Working closely with process chemists and control engineers to ensure the process plant is set up to provide maximum output levels and efficient running of the production facility.
- Designing plant and equipment configuration so that they can be readily adapted to suit the product range and the process technologies involved, considering environmental and economic aspects.
- Setting up scale-up and scale-down processes including appropriate changes to equipment design and configuration.
- Assessing options for plant expansion or reconfiguration by developing and testing process simulation models.
- Designing, installing and commissioning new production plants, including monitoring developments and troubleshooting.
- Optimising production by analysing processes and compiling de-bottleneck studies.
- Researching new products from trial through to commercialisation and improving product lines.
- Ensuring that potential safety issues related to the project operator, the environment, the process and the product are considered at all stages.

Skills

- An understanding of engineering principles and mathematics.
- An aptitude for and interest in chemistry.
- Project management skills.
- Resource management skills.
- Oral and written communication skills.
- Analytical and problem-solving ability.
- The ability to work as part of a team but the capacity to motivate and lead a team.
- Strong IT skills.
- A careful and methodical approach with good attention to detail.
- Commercial and business awareness.
- Creativity and innovation.



4a In your notebook, take the heading "Chemical Engineers - Knowledge, skills and roles" on a new page. Neatly copy the diagram above. Within each main branch of engineering, there are many specialists. Investigate if you can find more than the ones above and record what its specialities are.

4b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Chemical Engineers' Knowledge - What they know" and a list of "Electrical Engineers Skills - What they can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

Structural engineer role overview

Structural engineers design structures to withstand stresses and pressures imposed through environmental conditions and human use. They ensure buildings and other structures do not deflect, rotate, vibrate excessively or collapse and that they remain stable and secure throughout their use. They also examine existing buildings and structures to test if they are structurally sound and still fit for purpose. Working in close partnership with architects and other professional engineers they help to design most structures, including houses, hospitals, office blocks, bridges, oil rigs, ships and aircraft. It is the structural engineer's responsibility to choose the appropriate materials, such as concrete, steel, timber and masonry, to meet design specifications and they are often involved in inspecting the work and advising contractors.

Knowledge and responsibilities

- Analyse configurations of the basic structural components of a building or other structure.
- Calculate the pressures, stresses and strains that each component, such as a beam or lintel, will experience from other parts of the structure due to human use or environmental pressures such as weather or earthquakes.
- Consider the strength of various materials, e.g. timber, concrete, steel and brick, to see how their inclusion may necessitate a change of structural design.
- Liaise with other designers, including architects, to agree on safe designs and their fit with the aesthetic

concept of the construction.

- Examine structures at risk of collapse and advising how to improve their structural integrity, such as recommending removal or repair of defective parts or rebuilding the entire structure.
- Make drawings, specifications and computer models of structures for building contractors.
- Work with geotechnical engineers to investigate ground conditions and analyse results of soil sample and in-situ tests.
- Liaise with construction contractors to ensure that newly erected buildings are structurally sound.
- Apply expert knowledge of the forces that act on various structures.
- Use computers and computer-aided design (CAD) technology for simulation purposes.

Skills

- Good analytical and problem-solving skills.
- Strong mathematical ability.
- Computer literacy.
- A grasp of physics.
- Three-dimensional conceptual skills.
- Excellent oral and written communication skills.
- Diagrammatic skills.
- The ability to teamwork.
- Attention to detail.
- The ability to liaise well with professionals from other disciplines.
- An interest in the design and structure of buildings.

5a In your notebook, take the heading "Structural Engineers - Knowledge, skills and roles" on a new page. Within each main branch of engineering, there are many specialists. Investigate if you can find more than the ones above and record what its specialities are.

5b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Structural Engineers' Knowledge - What they know" and a list of "Structural Engineers Skills - What they can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

6a In your notebook, take the heading "Electronic Engineers - Knowledge, skills and roles" on a new page. Within each main branch of engineering, there are many specialists. Investigate if you can find more than the ones above and record what its specialities are.

6b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Electronic Engineers' Knowledge - What they know" and a list of "Electronic Engineers Skills - What they can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

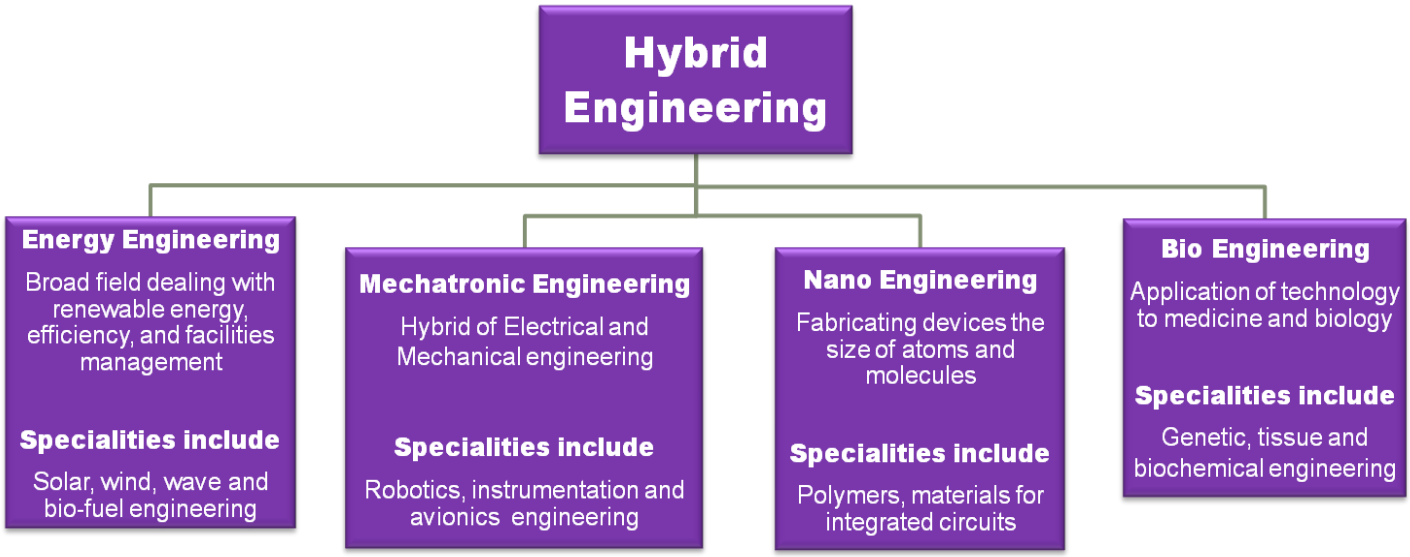
7a In your notebook, take the heading "Environmental Engineers - Knowledge, skills and roles" on a new page. Within each main branch of engineering, there are many specialists. Investigate if you can find more than the ones above and record what its specialities are.

7b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Environmental Engineers' Knowledge - What they know" and a list of "Environmental Engineers Skills - What they can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

Hybrid engineers (not part of your seven engineers)

Hybrid Engineers have become essential as Engineers have specialised more and more. This is because you often need someone who knows about several branches of Engineering 'A Jill or Jack of all trades' though they don't have to be a complete expert in them - just as a Doctor in General Practice (GP) may send you to a heart specialist, a chest specialist or an ear, nose and throat specialist.

For example, it is not uncommon to find that many civil engineers (especially structural engineers) know quite a lot about mechanical engineering, or vice versa, because often their work has a lot of overlapping knowledge and skills and they often work closely together.



8a In your notebook, take the heading "Hybrid engineers - Knowledge, skills and roles" on a new page. Neatly copy the diagram above. Within each main branch of engineering, there are many specialisms. Investigate if you can find more than the ones above and record what its specialities are.

8b In your notebook, research and describe a brief overview of roles and responsibilities for that Engineering type. You will also need to describe a list of "Hybrid Engineers' Knowledge - What they know" and a list of "Hybrid Engineers Skills - What the can do with their knowledge". You may use the statements above as a starting point, and research the internet for up to 10 minutes to help.

Other engineering:

Industrial engineering, Nuclear engineering, Agricultural engineering and applied engineering.

- Environmental engineers use the principles of engineering, soil science, biology, and chemistry to develop solutions to environmental problems. They are involved in efforts to improve recycling, waste disposal, public health, and water and air pollution control.

- Mechanical engineering is the discipline that applies the principles of engineering, physics, and materials science for the design, analysis, manufacturing, and maintenance of mechanical systems.

- Chemical engineering is a branch of engineering that applies physical sciences (physics and chemistry), life sciences (microbiology and biochemistry), together with applied mathematics and economics to produce, transform, transport, and properly use chemicals, materials and energy.

- Electrical engineering is a field of engineering that generally deals with the study and application of electricity, electronics, and electromagnetism.

- Electronics engineer is an electrical engineering discipline which utilizes non-linear and active electrical components (such as semiconductor devices, especially transistors, diodes and integrated circuits) to design electronic circuits, devices, microprocessors, microcontrollers and other systems.

- Computer Engineers develop and improve the software programs and hardware that make computers run. Computer Engineers may specialize in either software or hardware. From operating system software, such as Windows and Linux, to individual computer programs, such as Photoshop and Microsoft Office, Software Engineers turn piles of hardware into fully functional computers. Hardware Engineers develop the hardware of computers, including the motherboards, graphics and audio cards and drives that are later programmed by Software Engineers.

Impacts of Engineering

Within any engineering challenge or solution, Engineering can have a major impact on the World. Engineering has shaped the way people live and work. Engineering can have significant influence on the prosperity of people and can impact positively and sometimes negatively on the environment. Below is some exemplar statements which your case study report should contain.

Evidence of describing clearly the nature of the problem (within the case study report).

Critical analysis of wind energy as a solution to climate change

The need to conserve energy has become a very important issue in Scotland. Scientists have predicted that the planet will warm up and temperatures will rise by between 1.8°C and 4.0°C by the end of the century.

The reason for this rise in global temperature has been blamed on our burning of fossil fuels, especially for electricity generation. Burning fossil fuels leads to emissions of greenhouse gasses which appear to be a major factor in global warming, and governments around the World are now trying to cut emissions of greenhouse gasses.

The Scottish Government have set a target of 31% of electricity being generated by alternatives, by 2020. In fact in 2011, renewable electricity amounted to around 35% of Scottish consumption, beating the Scottish Government's target.

Without wind power and other renewable energy schemes the potential is that the planet could face disaster, some scientists suggest up to 30% of species could become extinct, sea level rises of 200–800mm (possibly up to 12m if ice sheets melt), threatening hundreds of millions of people with flooding, more drought in some parts and more extreme weather in others.

This description clearly outlines the problem in the candidate's own words, and the use of numerical data shows a suitable level of technical detail

Social

The most obvious influence engineering has had in recent times on the social aspects of people's lives is through information technology. The internet has allowed instant public access to massive amounts of information. Social media has allowed people a medium through which to create broadcast information for themselves.

Engineering allows 21st century humans far greater free time. Faster transport and road networks allows shorter journey times to and from work so more time can be spent at home. Less frustration stuck in traffic jams. In some cases, the internet allows workers to make their home their office.

Convenience in the home through technological innovation means more leisure time. Food can be stored for longer, power is generated for heat, lighting and cooking conveniently by a mains supply and entertainment can be had on demand from various sources.

Economic

Engineering can impact on the economic fortunes of individuals, businesses, large companies, even the whole country. Engineering is an important employer. Many people are employed in some form of engineering associated role in both the private and public sector companies.

Engineering projects can have far reaching economic influences. The large scale projects such as road building, rail infrastructure and construction developments can not only create jobs and training opportunities for the people involved but once completed can generate increased interest and investment in that part of the country. Bringing more employers and people, increasing wealth in the area. This can lead to increased population, creating a larger market for existing local businesses and retailers.

Evidence of describing some social and economic impacts of the solution (within the case study report).

Economic and social impacts can be shown to be both positive and negative.

On the positive side, wind turbines are being manufactured in some remote parts of Scotland, areas where jobs are quite hard to come by, this in turn leads to more money being available for the local economy, shops and businesses make increased profits and in turn employ more people, local school roles increase, and so on. Scotland's offshore wind industry looks set to deliver 28,000 jobs by 2020, contributing £7.1 billion to the economy

Many people however believe that wind farms in particular may put visitors off coming to areas where there are large scale wind farms. Tourism is a major industry in Scotland, Visit Scotland estimate that tourism brings £11 billion in to the country and pays the wages of 200,000 and tourism is Scotland's fourth largest export earner.

This description, in the candidate's own words, outlines both social and economic benefits and problems arising from wind power. Again, the inclusion of numerical data is appropriate at this level.

Environmental

Engineering can be a force for good or evil when looking at its impact on the environment. Engineering could be blamed for much of the environmental damage caused. Mechanised farming, industrial pollution, the car, the rise of cities can all be traced back to the industrial revolution of the early 19th century. However engineers of the 21st century realise the failings of past generations and engineering now seeks to address environmental issues and in some cases aims to correct the damage already done.

Engineering is an important factor in renewable power generation, recycling of materials, efficient use of energy, information distribution and many others. It is through engineering that communities in developing nations can better prosper and experience a better quality of life.

The environmental engineer focuses entirely on improving the impact of engineering on the Worlds environment.

Evidence of describing clearly some environmental impacts of the solution (within the case study report).

Generating more of our power by renewable energies will lead to many environmental impacts, some positive some negative.

The obvious positive impact will be the decrease in greenhouse gas emissions together with other polluting gasses, will clean up the atmosphere, and lead to the temperature and weather stabilising. Another less obvious one will be as nuclear stations are closed and decommissioned there will be less danger from the transportation of nuclear waste, and a smaller need for storage of this waste, although the waste we currently have is going to be with us for quite some time to come.

Negative impacts can be seen quite clearly, it is often difficult to look around in many parts of the country without seeing a major wind farm, to many people these wind farms are an eyesore, and many bodies such as the Ramblers oppose the building of new wind farms. The building or access roads may have a harmful impact on fragile ecosystems.

This description, in the candidate's own words, outlines both positive and negative impacts on the environment.

Emerging technologies

Emerging technologies are technologies that are perceived as capable of changing the status quo. Emerging technologies as of 2016 include a variety of technologies such as educational technology, information technology, nanotechnology, biotechnology, cognitive science, psychotechnology, robotics, and artificial intelligence. Graphene is a material that has the potential to revolutionise a huge number of areas from electronics to energy production and structures.

By searching online for 'Emerging Technologies', you will find lots of information out there. If you are struggling, the use of website 'Scientific America' and the 'World Economic Forum'. It creates a top ten list every year of emerging technologies. Another thing engineer's need to consider is how a new emerging technology or material that would be more suitable if used in a different way or product. Would the operation of a product by using an alternative fuel type for example make it more economical, environmentally friendly or efficient?

9 Research and create a data sheet on any emerging technology, process or material (up to five). Consider what is this technology, and how does it work? Also write a brief statement of how will the development of this change the World?

Name of material	What is the technology	How does it work?	How will the development change the World?

Outcome evidence: Identifying and describing emerging technologies which may impact future developments.

The candidate is required to provide brief descriptions of at least two emerging technologies which may impact on the chosen problem and/or solution. Emerging technologies is a broad term, and may encompass a wide range of areas, including new materials, intelligent systems, development and improvements to existing technologies (eg improved efficiency of PV panels), or developments which are still at the research (or even speculative) stage. These must clearly relate to the chosen problem and its solution.

10 Investigate two new emerging materials and write a 200+ word report giving details of its properties and how it could be used in products such mobile phones or cars, or in medical procedures etc.

CASE STUDY 1 - SCOTTISH ENGINEERING PROJECTS THE DUNDEE WATERFRONT REDEVELOPMENT

The existing waterfront area under the Tay Road Bridge had been suffering from ageing. Dundee City Council headquarters was based in a dilapidated tower block in the area that was generally considered to be an eyesore, the Olympia swimming pool was constantly needing repaired, and the Hilton hotel was very dated.

A decision was made to level the existing area, and turn it into a top class range of leisure and tourist attractions, including the rebuilding of the swimming pool, the building of the Victoria & Albert museum, renovation of the nearby rail station, improvement to the road traffic links, and the creation of green spaces.

Consider and research the Dundee Waterfront redevelopment project. Produce a 1000 word report covering the following points: A Powerpoint presentation is also suitable as a platform for the report.

(a) Give some context and challenges specific to the project. Why was the project needed? Images? (outcome 3.1)

(b) Do some research to identify an aspect of the project which requires the expertise of a mechanical engineer. List 3 skills and 3 areas of knowledge which a mechanical engineer would require for their role in **this** project.

(c) Do some research to identify an aspect of the project which requires the expertise of a civil engineer. List 3 skills and 3 areas of knowledge which a civil engineer would require for their role in **this** project.

(d) Do some research to identify an aspect of the project which requires the expertise of an electrical and electronic engineer? List 3 skills and 3 areas of knowledge which an electrical and electronic engineer would require for their role in **this** project.

(e) Do some research to identify an aspect of the project which requires the expertise of a chemical engineer? List 3 skills and 3 areas of knowledge which a chemical engineer would require for their role in **this** project.

(f) Describe one positive and one negative economic impact resulting from this project. (outcome 3.2)

(g) Describe one positive and one negative social impact resulting from an engineering project or challenge. (outcome 3.2)

(h) Describe one positive and one negative environmental impact resulting from this project. (outcome 3.3)

(i) Explain how emerging technologies were used (or could be used) within the project. (outcome 3.4)

CASE STUDY 2 - SCOTTISH ENGINEERING PROJECTS THE QUEENSFERRY CROSSING

The Queensferry Crossing was commissioned as the existing Forth Road Bridge had exceeded its intended design life. A Powerpoint presentation is also suitable as a platform for the report.

Over the near fifty years since it had been opened, the Forth Road Bridge was carrying many more vehicles between the M8 motorway, the Lothians and Fife than it had originally been designed for as there are many more vehicles on the road in the 21st century than there was in the 1960s.

Consider and research the Queensferry Crossing project. Produce a 1000 word report covering the following points: A Powerpoint presentation is also suitable as a platform for the report.

(a) Give some context and challenges specific to the project. Why was the project needed? Images? (outcome 3.1)

(b) Do some research to identify an aspect of the project which requires the expertise of a mechanical engineer. List 3 skills and 3 areas of knowledge which a mechanical engineer would require for their role in **this** project.

(c) Do some research to identify an aspect of the project which requires the expertise of a civil engineer. List 3 skills and 3 areas of knowledge which a civil engineer would require for their role in **this** project.

(d) Do some research to identify an aspect of the project which requires the expertise of an electrical and electronic engineer? List 3 skills and 3 areas of knowledge which an electrical and electronic engineer would require for their role in **this** project.

(e) Do some research to identify an aspect of the project which requires the expertise of a chemical engineer? List 3 skills and 3 areas of knowledge which a chemical engineer would require for their role in **this** project.

(f) Describe one positive and one negative economic impact resulting from this project. (outcome 3.2)

(g) Describe one positive and one negative social impact resulting from an engineering project or challenge. (outcome 3.2)

(h) Describe one positive and one negative environmental impact resulting from this project. (outcome 3.3)

(i) Explain how emerging technologies were used (or could be used) within the project. (outcome 3.4)

CASE STUDY 3 - SCOTTISH ENGINEERING PROJECTS THE BORDERS RAILWAY

Many railway lines connecting towns in the Scottish borders with each other and with Edinburgh had been shut down over decades with the decline of heavy industry as the industrial revolution came to an end in Scotland.

After a great deal on campaigning, an ambitious scheme to re-open the main line from Edinburgh Waverley through Newtongrange and on to many border towns was given the go-ahead, including the building or renovation of new stations, new bridges and sections of track.

Consider and research the Borders Railway project. Produce a 1000 word report covering the following points:

(a) Give some context and challenges specific to the project. Why was the project needed? Images? (outcome 3.1)

(b) Do some research to identify an aspect of the project which requires the expertise of a mechanical engineer. List 3 skills and 3 areas of knowledge which a mechanical engineer would require for their role in **this** project.

(c) Do some research to identify an aspect of the project which requires the expertise of a civil engineer. List 3 skills and 3 areas of knowledge which a civil engineer would require for their role in **this** project.

(d) Do some research to identify an aspect of the project which requires the expertise of an electrical and electronic engineer? List 3 skills and 3 areas of knowledge which an electrical and electronic engineer would require for their role in **this** project.

(e) Do some research to identify an aspect of the project which requires the expertise of a chemical engineer? List 3 skills and 3 areas of knowledge which a chemical engineer would require for their role in **this** project.

(f) Describe one positive and one negative economic impact resulting from this project. (outcome 3.2)

(g) Describe one positive and one negative social impact resulting from an engineering project or challenge. (outcome 3.2)

(h) Describe one positive and one negative environmental impact resulting from this project. (outcome 3.3)

(i) Explain how emerging technologies were used (or could be used) within the project. (outcome 3.4)